

Problem Set 1: Answers

Probability and Statistical Inference

Tom Glinnan

1. **Identification and estimation.** A voluntary study-skills program is evaluated using final examination scores. Let $D_i = 1$ if student i participates in the program and $D_i = 0$ otherwise. Let Y_i be the observed score. The sample is

D_i	1	1	1	1	0	0	0	0
Y_i	78	74	82	86	65	72	68	71

Let

$$\hat{\beta} = \bar{Y}_1 - \bar{Y}_0, \quad \beta = \mathbb{E}[Y \mid D = 1] - \mathbb{E}[Y \mid D = 0].$$

- (a) Compute $n_1, n_0, \bar{Y}_1, \bar{Y}_0$ and $\hat{\beta}$. Then identify the estimator, its numerical estimate in this sample, and the parameter.
- (b) Introduce potential outcomes $Y(1)$ and $Y(0)$ and write the equation that links them to the observed outcome Y . Define the ATET.
- (c) Show that

$$\beta = \text{ATET} + [\mathbb{E}[Y(0) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 0]].$$

Give an interpretation of the term in square brackets.

- (d) In the population, suppose

$$\mathbb{E}[Y(1) \mid D = 1] = 78, \quad \mathbb{E}[Y(0) \mid D = 1] = 72, \quad \mathbb{E}[Y(0) \mid D = 0] = 68.$$

Compute β , the selection-bias term, and the ATET. Does the raw population difference overstate or understate the ATET?

- (e) Now suppose that program participation were randomly assigned, so that $D \perp\!\!\!\perp (Y(0), Y(1))$. What additional conclusions would follow? Explain why a very large sample, by itself, cannot correct a failure of this identification assumption.

Solution.

- (a) There are four observations in each group, so $n_1 = n_0 = 4$. The group totals are

$$78 + 74 + 82 + 86 = 320, \quad 65 + 72 + 68 + 71 = 276.$$

Consequently

$$\bar{Y}_1 = 320/4 = 80, \quad \bar{Y}_0 = 276/4 = 69, \quad \hat{\beta} = 80 - 69 = 11.$$

Before the sample is observed, $\hat{\beta}$ denotes the random rule and is the *estimator*. Its realized value, 11, is the *estimate*. The fixed population comparison β is the *parameter*. Whether β is also a causal parameter requires an identification argument.

- (b) $Y(1)$ is the score the student would have under participation and $Y(0)$ is the score the same student would have without participation. The observation rule is

$$Y = DY(1) + (1 - D)Y(0).$$

The average treatment effect on the treated is

$$\text{ATET} = \mathbb{E}[Y(1) - Y(0) \mid D = 1].$$

- (c) By the observation rule, $\mathbb{E}[Y \mid D = 1] = \mathbb{E}[Y(1) \mid D = 1]$ and $\mathbb{E}[Y \mid D = 0] = \mathbb{E}[Y(0) \mid D = 0]$. Therefore

$$\begin{aligned}\beta &= \mathbb{E}[Y(1) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 0] \\ &= \mathbb{E}[Y(1) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 1] \\ &\quad + \mathbb{E}[Y(0) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 0] \\ &= \text{ATET} + [\mathbb{E}[Y(0) \mid D = 1] - \mathbb{E}[Y(0) \mid D = 0]].\end{aligned}$$

The term in square brackets compares the two groups' mean outcomes in the no-program state. It is selection bias: a difference that would have existed even if nobody had received the program.

- (d) The observable population comparison is

$$\beta = 78 - 68 = 10.$$

The selection-bias term is $72 - 68 = 4$, while

$$\text{ATET} = 78 - 72 = 6.$$

Thus $10 = 6 + 4$. Positive selection bias makes the raw difference overstate the ATET by four points.

- (e) Random assignment gives

$$\mathbb{E}[Y(d) \mid D = 1] = \mathbb{E}[Y(d) \mid D = 0] = \mathbb{E}[Y(d)], \quad d = 0, 1.$$

It follows that

$$\beta = \mathbb{E}[Y(1)] - \mathbb{E}[Y(0)] = \text{ATE}.$$

It also makes $\text{ATET} = \text{ATE}$. A larger sample can make $\hat{\beta}$ very close to the descriptive population difference β ; that is an estimation result. If the selection term is actually nonzero, then β differs from the ATET, and increasing n cannot remove that discrepancy. Without random assignment, causal identification is no longer guaranteed; it may still follow from equality of the groups' mean untreated potential outcomes or another valid research design.

2. Probability rules and updating. You may use the probability axioms: non-negativity, $\mathbb{P}(\Omega) = 1$, and countable additivity for disjoint events.

- Prove that $\mathbb{P}(A^c) = 1 - \mathbb{P}(A)$.
- If $A \subseteq B$, prove that $\mathbb{P}(A) \leq \mathbb{P}(B)$.
- Show that $\mathbb{P}(A \cup B) \leq \mathbb{P}(A) + \mathbb{P}(B)$.
- A data-quality problem is present with prior probability 0.30. An automatic check raises a warning with probability 0.80 when the problem is present and with probability 0.20 when it is absent. Find the probability that the problem is present after a warning is observed. Identify the prior, likelihood and posterior in your calculation, and explain why $\mathbb{P}(\text{warning} \mid \text{problem})$ is not the requested posterior.

Solution. Write H for “problem” and W for “warning.”

- (a) The events A and A^c are disjoint and their union is Ω . Additivity and total mass give

$$1 = \mathbb{P}(\Omega) = \mathbb{P}(A) + \mathbb{P}(A^c),$$

which rearranges to the required formula.

- (b) When $A \subseteq B$, the disjoint union $B = A \cup (B \setminus A)$ gives

$$\mathbb{P}(B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \geq \mathbb{P}(A),$$

where the inequality follows from non-negativity.

- (c) The union can be written as the disjoint union $A \cup (B \setminus A)$. Hence

$$\mathbb{P}(A \cup B) = \mathbb{P}(A) + \mathbb{P}(B \setminus A) \leq \mathbb{P}(A) + \mathbb{P}(B),$$

using part (b), since $B \setminus A \subseteq B$.

- (d) The prior is $\mathbb{P}(H) = 0.30$ and the likelihood of the observed warning under H is $\mathbb{P}(W | H) = 0.80$. The probability of a warning is

$$\begin{aligned}\mathbb{P}(W) &= \mathbb{P}(W | H)\mathbb{P}(H) + \mathbb{P}(W | H^c)\mathbb{P}(H^c) \\ &= 0.80(0.30) + 0.20(0.70) = 0.24 + 0.14 = 0.38.\end{aligned}$$

Bayes' rule therefore gives the posterior

$$\mathbb{P}(H | W) = \frac{\mathbb{P}(W | H)\mathbb{P}(H)}{\mathbb{P}(W)} = \frac{0.24}{0.38} = \frac{12}{19} \approx 0.632.$$

$\mathbb{P}(W | H) = 0.80$ conditions on the problem being present. The question instead conditions on the warning having been observed, so the conditioning event and the normalizing denominator are different.

3. **An exact sampling distribution.** Each observation is equally likely to be 0 or 2, and the population mean is $\theta = 1$. For a sample of size two, the four ordered samples (0, 0), (0, 2), (2, 0), (2, 2) each have probability 1/4. Consider $T_2 = (X_1 + X_2)/2$ and $S_2 = X_1$.

- (a) Calculate both estimators in each possible sample. Hence give their sampling distributions: their possible values and the probability of each value.
- (b) You are given that both sampling means equal 1, while $\text{Var}(T_2) = 1/2$ and $\text{Var}(S_2) = 1$. Calculate their biases and MSEs. Which is more efficient within this pair of unbiased estimators?
- (c) For independent samples of size n , define $T_n = \bar{X}_n$ and $S_n = X_1$. Their sampling means and variances are given below:

	mean	variance
T_n	1	$1/n$
S_n	1	1

Describe what happens to the centre and spread of each sampling distribution as n grows. Explain directly why S_n is not consistent for $\theta = 1$.

- (d) Explain the difference between increasing the sample size n and increasing the number R of repeated samples used to draw a histogram of T_n . Which one changes the sampling distribution of the estimator?

Solution.

- (a) The four equally likely samples give

(X_1, X_2)	(0, 0)	(0, 2)	(2, 0)	(2, 2)
T_2	0	1	1	2
S_2	0	0	2	2

Adding the probabilities of samples that give the same estimate,

$$\Pr(T_2 = 0) = \frac{1}{4}, \quad \Pr(T_2 = 1) = \frac{1}{2}, \quad \Pr(T_2 = 2) = \frac{1}{4}.$$

For S_2 , $\Pr(S_2 = 0) = \Pr(S_2 = 2) = 1/2$.

- (b) Bias is the sampling mean minus the target, so both biases are zero. Using Lecture 1's identity $\text{MSE} = \text{Bias}^2 + \text{Var}$,

$$\text{MSE}(T_2) = \frac{1}{2}, \quad \text{MSE}(S_2) = 1.$$

Both are unbiased; T_2 is more efficient within this pair because its variance is smaller.

- (c) Both centres remain at the target 1. The variance of T_n decreases as $1/n$, while the spread of S_n stays unchanged. Indeed, S_n is always either 0 or 2, so $|S_n - 1| = 1$. For example,

$$\Pr(|S_n - 1| > 1/2) = 1$$

for every n . This violates the definition of consistency.

- (d) Increasing n changes T_n and its sampling distribution, including its variance $1/n$. Increasing R at fixed n gives more repetitions of the same experiment. The histogram becomes a better picture of the same distribution; the estimator's variance does not change.

4. MSE and consistency. Use Lecture 1's identity $\text{MSE} = \text{Bias}^2 + \text{Var}$ and its definition of consistency.

- (a) An estimator T equals $\theta - 1$ or $\theta + 3$, each with probability $1/2$. Its sampling mean is $\theta + 1$ and its variance is 4. Calculate its bias and MSE. Check the MSE by averaging the squared errors in these two equally likely cases.
- (b) An estimator V_n equals $\theta - 1/n$ or $\theta + 1/n$, each with probability $1/2$. Its sampling mean is θ and its variance is $1/n^2$. Calculate its bias and MSE. Then use the definition of consistency directly to show that $V_n \xrightarrow{p} \theta$.
- (c) Let $W_n = n$ with probability $1/n$ and $W_n = 0$ otherwise, viewed as an estimator of zero. You are given that its bias is 1 and its variance is $n - 1$. Show directly that $W_n \xrightarrow{p} 0$. What does this example tell us about whether consistency requires bias and variance to converge to zero?

Solution.

- (a) The bias is $(\theta + 1) - \theta = 1$, and $\text{MSE}(T) = 1^2 + 4 = 5$. The two possible errors are -1 and 3 , so averaging their squares gives

$$\frac{(-1)^2 + 3^2}{2} = 5,$$

as required.

- (b) The bias is zero and the MSE is $1/n^2$. Every possible estimate is exactly $1/n$ from θ . Fix any $\varepsilon > 0$. Once $n > 1/\varepsilon$,

$$\Pr(|V_n - \theta| > \varepsilon) = 0.$$

This probability therefore converges to zero, which is the definition of $V_n \xrightarrow{p} \theta$.

- (c) Fix any $\varepsilon > 0$. For all $n > \varepsilon$, the only value more than ε from zero is n , so

$$\Pr(|W_n| > \varepsilon) = \Pr(W_n = n) = \frac{1}{n} \longrightarrow 0.$$

Thus W_n is consistent for zero, although its bias stays at 1 and its variance grows. Its MSE is $1 + (n - 1) = n$. Consistency concerns the probability of being far from the target: here, large mistakes become rarer even as their size increases. Vanishing bias and variance are not necessary for consistency.

5. **Four estimators.** Consider the following version of Lecture 1's normal sampling examples:

$$A_n = X_1, \quad B_n = \bar{X}_n, \quad C_n = \bar{X}_n + \frac{2}{n}, \quad D_n = 0.$$

Their sampling distributions are given; you do not need to derive them. Here $N(m, v)$ denotes a normal distribution with mean m and variance v .

estimator	sampling distribution
A_n	$N(\theta, 4)$
B_n	$N(\theta, 4/n)$
C_n	$N(\theta + 2/n, 4/n)$
D_n	always equals 0

- For each estimator, calculate its bias, variance and MSE.
- Which estimators are unbiased, and which are consistent for a general unknown θ ? Explain consistency using the centres and spreads of the normal curves, as in Lecture 1.
- Compare A_n and B_n within the class of unbiased estimators. Why does D_n show that smallest variance is not a useful definition of efficiency unless the target and the set of estimators being compared are specified?
- Which estimator illustrates that an estimator may be biased at every finite n and nevertheless be consistent?

Solution.

- Read the means and variances from the given distributions, subtract θ to obtain each bias, and use $\text{MSE} = \text{Bias}^2 + \text{Var}$:

	bias	variance	MSE
A_n	0	4	4
B_n	0	$4/n$	$4/n$
C_n	$2/n$	$4/n$	$4/n + 4/n^2$
D_n	$-\theta$	0	θ^2

- A_n and B_n are unbiased. C_n is biased at every finite n . D_n is unbiased only when $\theta = 0$.
The normal curve for B_n stays centred at θ and gets narrower. For C_n , its centre approaches θ and its spread shrinks too. As in the lecture's normal examples, both distributions concentrate around θ , so both estimators are consistent. The curve for A_n never changes: it remains $N(\theta, 4)$, so it does not concentrate and is not consistent. D_n stays at zero and is consistent only in the special case $\theta = 0$.
- For $n > 1$, $4/n < 4$, so B_n is more efficient within this pair of unbiased estimators. D_n has zero variance but can be far from the target; its MSE is θ^2 . We must specify both the target and a comparison class, such as unbiased estimators.
- C_n : its bias is $2/n \neq 0$ for every finite n , yet its normal sampling distribution concentrates around θ .

6. **Sample summaries and interpretation.** Consider the ordered sample

1, 2, 3, 4, 5, 6, 14.

Use the lecture convention $\hat{q}_p = X_{(\lceil np \rceil)}$.

- (a) Compute the sample mean, median, first and third quartiles, and interquartile range.
- (b) Replace the largest observation, 14, by 70 and recompute these summaries. Which measures are most sensitive to this change?
- (c) State whether each claim is true or false and explain briefly.
 - i. Increasing the number of simulation repetitions R reduces the variance of the estimator whose sampling distribution is being simulated.
 - ii. An estimator can be biased for every finite sample size and still be consistent.
 - iii. An estimator can have small MSE even though its realized estimate in one particular sample is far from the target.
 - iv. If $\hat{\beta}$ estimates β extremely precisely, then β must equal the causal effect of interest.

Solution.

- (a) Here $n = 7$ and the sum is 35, so $\bar{X} = 35/7 = 5$. The middle observation is $X_{(4)} = 4$, hence the median is 4. The stated convention gives

$$\hat{q}_{.25} = X_{(\lceil 7/4 \rceil)} = X_{(2)} = 2, \quad \hat{q}_{.75} = X_{(\lceil 21/4 \rceil)} = X_{(6)} = 6.$$

Thus the interquartile range is $6 - 2 = 4$.

- (b) The new sum is $35 - 14 + 70 = 91$, so the new mean is $91/7 = 13$. The ordered sample is now 1, 2, 3, 4, 5, 6, 70, so the median remains 4, the quartiles remain 2 and 6, and the IQR remains 4. The mean is highly sensitive to the extreme upper observation; these order-based central summaries are unchanged in this example.
- (c)
 - i. **False.** At fixed n , more repetitions make the simulation histogram more accurate. They do not change the estimator or its sampling variance.
 - ii. **True.** For example, C_n in Question 5 has bias $2/n \neq 0$ at every finite n but is consistent because its bias and variance both vanish.
 - iii. **True.** MSE is an average squared error across repeated samples. A low-MSE estimator can still make a large error in an unusual particular sample; low MSE is not a guarantee of small error in every realization.
 - iv. **False.** A small standard error concerns estimation of the population parameter β . Equality of β with a causal target is a separate identification claim and requires assumptions such as random assignment.